

Chen Qian

GitHub: [Chen20191020](#) | Website: [chen20191020.github.io](#)
Davis, California | mrcqian@ucdavis.edu | +1 571-267-0414

Summary

- **Applied Mathematics Ph.D. candidate at UC Davis** with an **M.S. in Statistics** and a focus on time series analysis, machine learning for sequential data, and statistical inference.
- Over 4 years of research experience in **sequential-data methods**, with applications to biomedical signal analysis and user behavior modeling.
- **First author** of a paper published in a **top-tier statistics journal (JRSSB)** on structural classification of locally stationary time series; applied the proposed method to clinical EEG classification.

Education

University of California, Davis <i>Ph.D. in Applied Mathematics</i>	Expected 2028 <i>Davis, CA</i>
University of California, Davis <i>M.S. in Statistics</i>	June 2023 <i>Davis, CA</i>
Complutense University of Madrid <i>B.S. in Applied Statistics</i>	July 2020 <i>Madrid, Spain</i>

Selected Research Projects

Structural Classification of Locally Stationary Time Series	Nov. 2024 – Apr. 2026
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- Developed an interpretable classification framework for sequential data using features derived from time-varying second-order dependence.
- Estimated time-varying autoregressive structure through sieve-based approximation and summarized local versus global temporal variation for classification.
- Evaluated the method on brain signal data and benchmarked it against 10 competing approaches, including convolution methods, shapelet methods, random forests, and recurrent neural networks.
- Achieved up to **94.7% test accuracy** on abnormal-versus-normal brain signal classification, outperforming competitive statistical and deep learning baselines.

Time-frequency analysis for non-stationary signals	Apr. 2025 – June 2025
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- Converted raw financial price and brain signals into time-frequency features that capture how signal patterns change over time and frequency, using PWVD and WPT.
- Built and evaluated machine-learning models to predict future price direction across 26 financial assets and to classify brain signals as normal or abnormal.
- Achieved **74.7% average price-direction accuracy** under the best financial-model setting and **81.3% brain signals classification accuracy**; also found that WPT was about **33 times faster** than PWVD, while PWVD provided slightly higher average prediction accuracy.

Simultaneous Inference for Locally Stationary Time Series	Sept. 2025 – Present
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- Developing estimators for autocorrelation function in locally stationary time series, where dependence evolves continuously over time.
- Establishing consistency and convergence guarantees under temporally dependent data, providing a reliable framework for tracking evolving dependence patterns across different lag values.
- Implementing Gaussian multiplier-bootstrap confidence bands to quantify uncertainty over the full time domain for estimated autocorrelation function.

Publication

- **Qian, C., Ding, X., & Li, L. (2026).** Structural classification of locally stationary time series based on second-order characteristics. **Journal of the Royal Statistical Society Series B: Statistical Methodology**, qkag083.

Professional Experience

Zhongyu Tech

Sept. 2020 – May 2021

Data Scientist Intern, Mobile Game Analytics

Wuhan, China

- Designed product experiments and analyzed monetization features for the casual mobile game *Wheel Offroad 3D*.
- Conducted cohort, retention, and churn analyses to identify player-engagement patterns and support product-growth decisions.
- Built ETL workflows and recurring analytical reports, translating gameplay data into actionable recommendations for product and engineering teams.

Technical Skills

Programming: Python, R, SQL

Languages: Mandarin Chinese (native), English (professional), Spanish (working proficiency).

Teaching & Leadership

University of California, Davis

July 2026

COSMOS Teaching Fellow, Exploring Data Science and Big Data Innovation

Davis, CA

- Instructed academically advanced high-school students in an intensive summer program centered on real-world data analysis.
- Mentored project teams in exploratory analysis, visualization, statistical modeling, scientific communication, and data-driven decision making.

University of California, Davis

2023 – Present

Teaching Assistant

Davis, CA

- Supported courses in statistics, data science, probability, and calculus through discussion sections, office hours, grading, and exam administration.

Honors & Awards

- 2026 L&S Graduate Summer Support Award, University of California, Davis (selected; declined due to COSMOS Teaching Fellow appointment).